

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer at an early-stage healthtech startup, building full-stack product (Next.js + Firebase/GCP) for a provider-facing **SMART on FHIR** app that surfaces covered, guideline-backed medication alternatives inside the EHR prescribing flow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend that builds alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR (Contextualized Construct Representations) text-analysis platform (Python, **FastAPI**, sentence-transformers, React) end to end, owning architecture and technical decisions while productionizing a published NLP method into a self-serve web tool for researchers.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system; grew the backend suite to **40 tests** (hermetic CI).

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant eTriage service stable.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a cross-platform desktop app with a fully offline, on-device pipeline (MediaPipe hand tracking, whisper.cpp ASR, local TTS) on a FastAPI + Electron/React stack, designing async APIs and queues to keep the UI responsive; shipped end to end under time pressure at the Qualcomm Edge AI Hackathon 2025.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end Spark ETL runtime **~25%** (17–19 min down to 14.3 min) on a **~1.4B-row** NYC TLC dataset via broadcast joins, AQE, skew-join handling, and shuffle-partition and column-pruning tuning, profiling stage-level timings in Spark UI.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Bash

Backend & Web: Node.js, Express, FastAPI, React, Next.js, REST APIs, WebSockets

Databases: MongoDB, PostgreSQL, Redis, SQLite

Cloud & DevOps: AWS, GCP, Azure, Docker, Git, GitHub Actions, Kafka, Kubernetes

ML & AI: PyTorch, sentence-transformers, Hugging Face, LLM agents, RAG, evaluation

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**.

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.